Enterprise Airline Analytics - Federated Insights

Metric Definitions & Semantic Registry	Centralized KPI reference detailing definitions, calculation logic, grain alignment, and certification status to ensure reporting transparency.
Executive Overview	Enterprise performance pulse providing consolidated visibility into profitability, demand trajectory, operational stability, and strategic risk exposure.
Route & Network Analytics	Diagnostic view of route-level profitability, volatility, and structural risk supporting network planning and capacity optimization decisions.
Customer & Revenue Segmentation	Segment-level revenue quality and booking behavior analysis linking demand predictability to enterprise performance outcomes.
Operational Insights & Execution Stability	Execution control layer evaluating reliability trends, disruption concentration, and financial impact of operational instability.
KPI Governance & COE	Governance oversight framework surfacing metric certification coverage, ownership accountability, and semantic alignment across domains.

METRIC DEFINITIONS & SEMANTIC REGISTRY

KPI	Definition	Category	Fact Source	Owner	Data Steward	Source Table	Calculation Logic
% Certified KPIs	Percentage of KPIs certified	Governance	Dim_KPI_Definitions	Head of CoE	Lead CoE Analyst	Dim_KPI_Definitions	Certified KPI Count / Total KPIs
Advance Booking Window (Days)	Average number of days between booking and flight date	Transformational	Fact_Bookings	Head of Commercial	Lead Commercial Analyst	Fact_Bookings	AVERAGE(Booking_Lead_Days)
Ancillary Revenue %	Percentage of revenue from ancillary services	Commercial	Fact_Bookings	Head of Commercial	Lead Commercial Analyst	Fact_Bookings	Ancillary Revenue / Total Revenue
Cancellation Rate %	Cancelled flights divided by total flights	Operational	Fact_Flights	Head of Operations	Lead Operations Analyst	Fact_Flights	Cancelled Flights / Total Flights
Certified KPI Count	Total number of KPIs marked as certified	Governance	Dim_KPI_Definitions	Head of CoE	Lead CoE Analyst	Dim_KPI_Definitions	COUNT where Certified_Flag = Yes
Cost per Route	Average monthly operating cost per route	Financial	Fact_Financials	Head of Finance	Lead Finance Analyst	Fact_Financials	Total_Cost by Route
Demand Momentum Index	3-month rolling growth rate in tickets sold	Transformational	Fact_Bookings	Head of Commercial	Lead Commercial Analyst	Fact_Bookings	Rolling 3M Tickets vs Previous 3M Ti
Fuel Cost Ratio	Fuel cost as percentage of total operating cost	Financial	Fact_Financials	Head of Finance	Lead Finance Analyst	Fact_Financials	Fuel_Cost / Total_Cost
Load Factor %	Seats sold divided by total seat capacity	Operational	Fact_Flights	Head of Operations	Lead Operations Analyst	Fact_Flights	Seats_Sold / Seats_Capacity
On-Time Departure %	Flights departing within 15 minutes of schedule	Operational	Fact_Flights	Head of Operations	Lead Operations Analyst	Fact_Flights	On-Time Flights / Total Flights
Premium Cabin Revenue %	Revenue share from premium cabin classes	Commercial	Fact_Bookings	Head of Commercial	Lead Commercial Analyst	Fact_Bookings	Premium Revenue / Total Revenue
Profit Margin %	Route profit divided by total revenue	Executive	Composite	Head of Finance	Lead Finance Analyst	Composite	Route Profit / Total Revenue
Revenue Forecast Accuracy %	Accuracy of revenue forecast relative to actuals	Transformational	Fact_Bookings	Head of CoE	Lead CoE Analyst	Fact_Bookings	1 - ABS((Actual Revenue - Forecast R Forecast Revenue)
Revenue per Flight	Average revenue generated per flight	Commercial	Composite	Head of Commercial	Lead Commercial Analyst	Composite	Total Revenue / Distinct Flights
Route Profit	Revenue minus total operating cost	Executive	Composite	Head of Finance	Lead Finance Analyst	Composite	[Total Revenue] - [Total Operating Cc
Route Risk Score	Composite score combining load volatility, margin volatility and cancellation rate	Transformational	Composite	Head of CoE	Lead CoE Analyst	Composite	Standardized composite risk index
Ticket Forecast Accuracy %	Accuracy of ticket forecast relative to actuals	Transformational	Fact_Bookings	Head of CoE	Lead CoE Analyst	Fact_Bookings	1 - ABS((Actual Tickets - Forecast Tic Forecast Tickets)
Total Operating Cost	Total operational cost including fuel, staff and airport fees	Executive	Fact_Financials	Head of Finance	Lead Finance Analyst	Fact_Financials	SUM(Fact_Financials[Total_Cost])
Total Passengers	Total tickets sold across routes	Executive	Fact_Bookings	Head of Commercial	Lead Commercial Analyst	Fact_Bookings	SUM(Fact_Bookings[Tickets_Sold])

For questions & inquiries, contact Anil Jacob at Aniljacobs@gmail.com, or visit my GitHub
This project repo is available at the following link: https://github.com/AwesomeAnil/Federated-Insights-Case-Study

EXECUTIVE OVERVIEW - YEAR 2024

Close Panel

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Select a Year:

- ○ 2022
- ○ 2023
- ◉ 2024

Select a route:

- ☐ DXB-BOM
- ☐ DXB-JFK
- ☐ DXB-LHR
- ☐ DXB-SIN
- ☐ DXB-SYD

Select a fare class:

- ☐ Premium
  - ☐ Premium Eco.
  - ☐ Business
  - ☐ First
- ☐ Standard

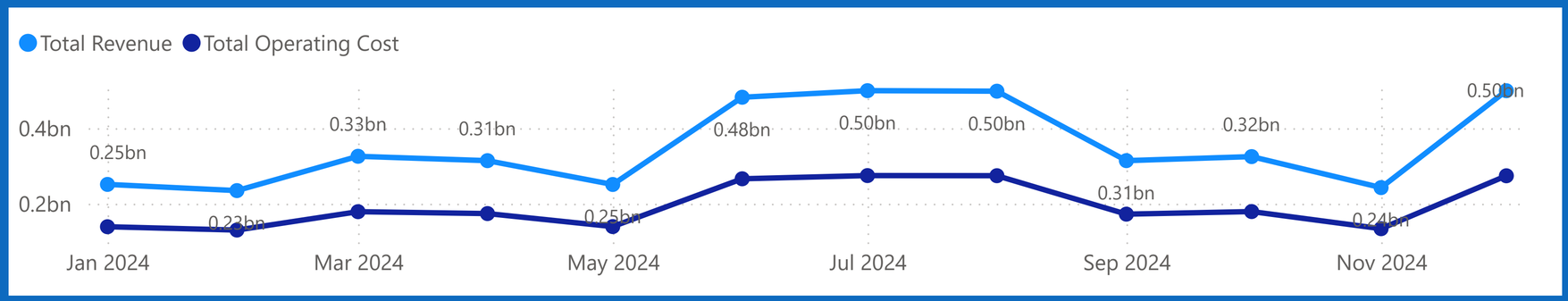
Select a segment:

- ☐ Business Traveller
- ☐ Corporate Contract
- ☐ Leisure Traveller
- ☐ Loyalty Gold

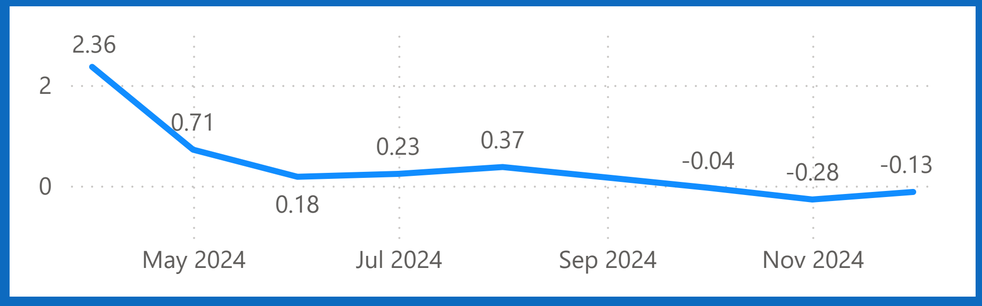
Executive KPIs

4.2bn	1.9bn	81.5%	89.7%	-0.13
Total Revenue	Route Profit	Profit Margin %	Revenue Forecast A...	Demand Momentu...

Financial Performance & Trends



Demand & Operational Signals



81.7%	38.5%
Load Factor %	On-Time Departure %
4.3%	**71.83**
Cancellation Rate %	Avg Tickets Sold

Risk & Exposure

Route_ID	Route Risk Score	Profit Margin %	Load Factor %
DXB-LHR	1.50	81.2%	81.7%
DXB-JFK	1.48	82.0%	81.8%
DXB-BOM	1.45	81.1%	81.8%
DXB-SIN	1.31	81.5%	81.3%
Total	**1.04**	**81.5%**	**81.7%**

0.31	1.04
Fuel Cost Ratio	Route Risk Score

Executive KPI's

Consolidated view of revenue growth, profitability, forecast reliability, and forward demand momentum to assess overall business health and trajectory.

Financial Performance Trends:

Illustrates revenue progression, operating cost pressure, and margin resilience across seasonal demand cycles and fuel volatility impacts.

Demand and Operational Signals:

Surfaces booking acceleration trends and operational stability indicators that influence near-term revenue realization and customer reliability.

Risk & Exposure:

Highlights route-level volatility, margin sensitivity, and operational disruption risk to support proactive network and cost management decisions.

ROUTE ANALYTICS (NETWORK PLANNING) - YEAR 2024

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Select a Year:

- 2022
- 2023
- 2024

Select a route:

- DXB-BOM
- DXB-JFK
- DXB-LHR
- DXB-SIN
- DXB-SYD

Select a fare class:

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  - Premium Eco.
  - Business
  - First
- Standard

Select a segment:

- Business Traveller
- Corporate Contract
- Leisure Traveller
- Loyalty Gold

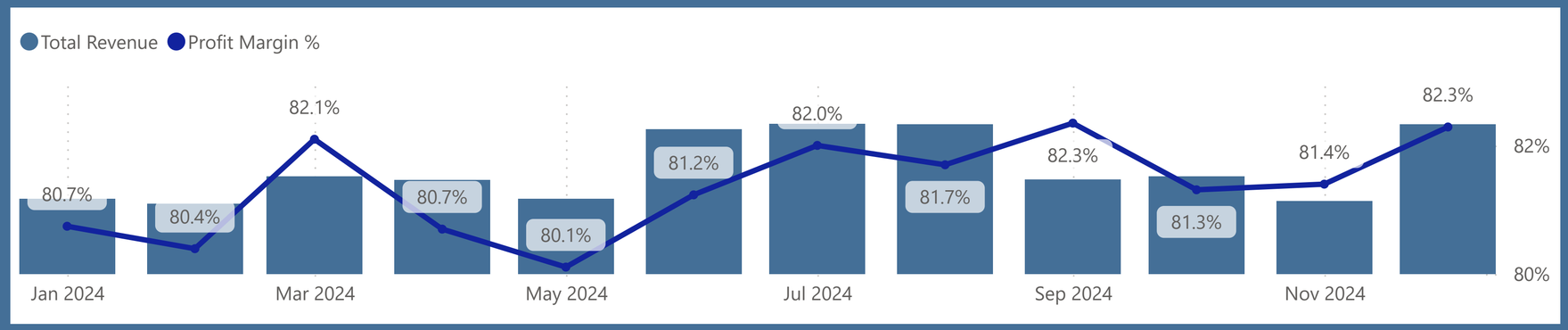
Route Health Snapshot

4.2bn	81.5%	81.7%	1.04	194.40M
Total Revenue	Profit Margin %	Load Factor %	Route Risk Score	Cost per Route

Route Health Snapshot

Provides a consolidated health view of the selected route, combining profitability, demand strength, and structural risk to assess strategic viability.

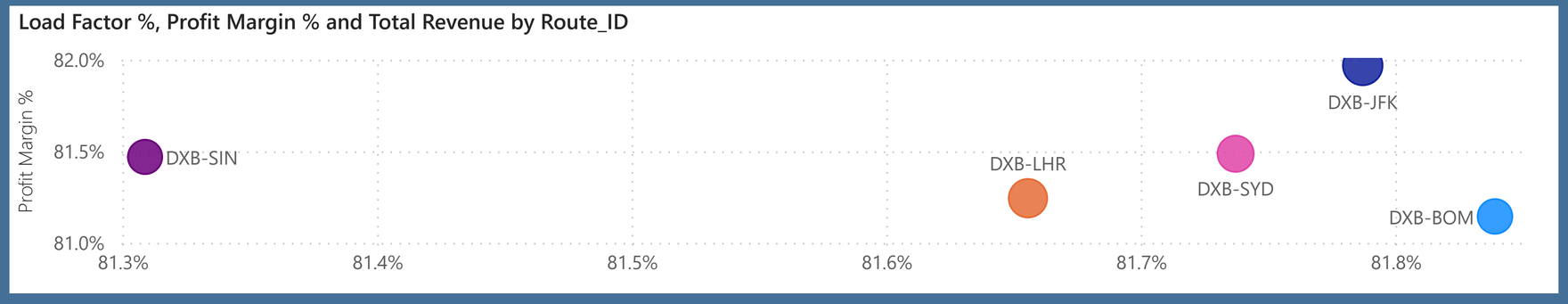
Route Revenue & Margin Trend



Route Revenue & Margin Trend

Reveals whether profitability is improving, deteriorating, or structurally constrained by cost pressure.

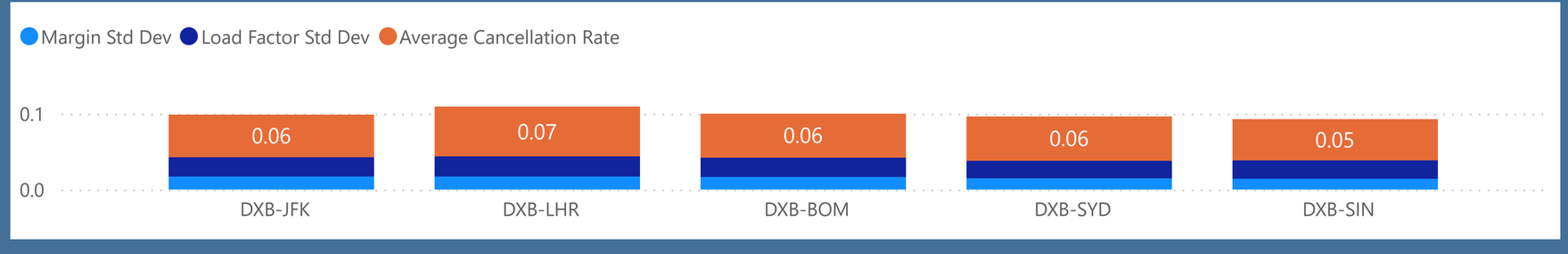
Performance Quadrant



Performance Quadrant

Categorizes network performance by load strength and margin resilience to guide capacity and pricing decisions.

Risk & Volatility breakdown



Risk & Volatility

Deconstructs the volatility drivers influencing route-level risk exposure across margin stability, demand variability, and operational disruption.

CUSTOMER & COMMERCIAL SEGMENTATION- YEAR 2024

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Select a Year:

- 2022
- 2023
- 2024

Select a route:

- DXB-BOM
- DXB-JFK
- DXB-LHR
- DXB-SIN
- DXB-SYD

Select a fare class:

- Premium
  - Premium Eco.
  - Business
  - First
- Standard

Select a segment:

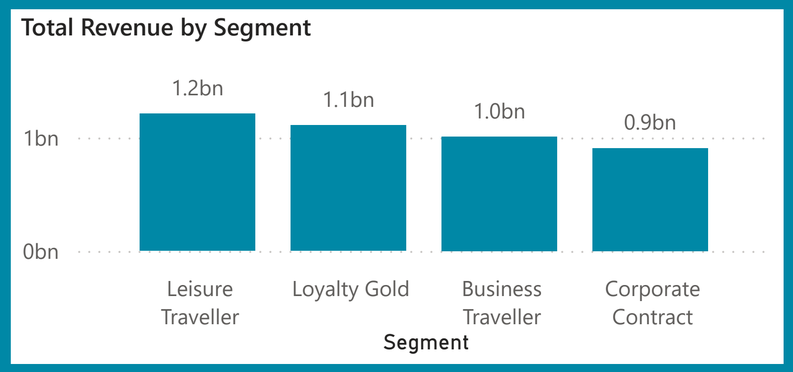
- Business Traveller
- Corporate Contract
- Leisure Traveller
- Loyalty Gold

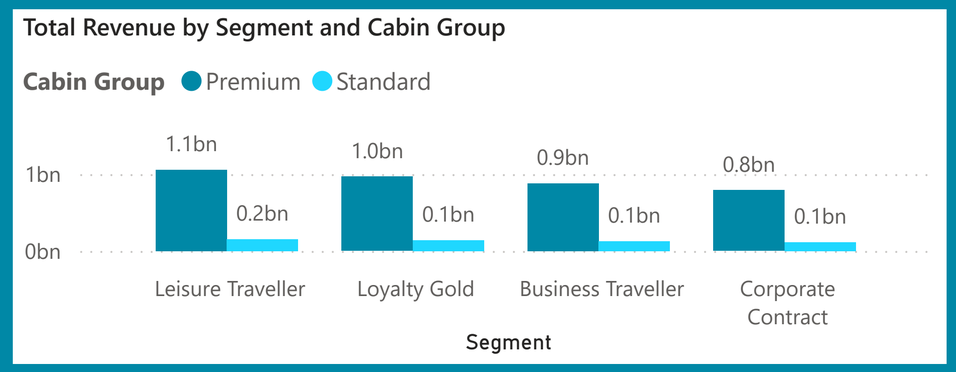
Segment Performance Snapshot

4.2bn	2,013	87.3%	2M	44.8
Total Revenue	Yield per Passenger	Premium Cabin Reve...	Total Passengers	Advance Booking Wi...

Segment Performance Snapshot:
Consolidates segment-level revenue contribution, yield quality, and booking behavior to assess structural demand value.

Revenue & Mix Analysis

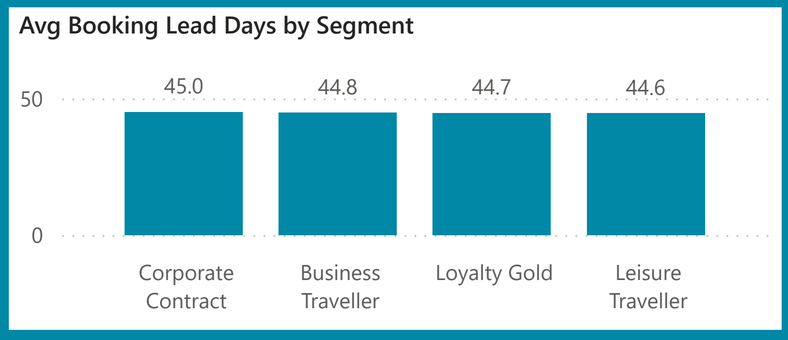


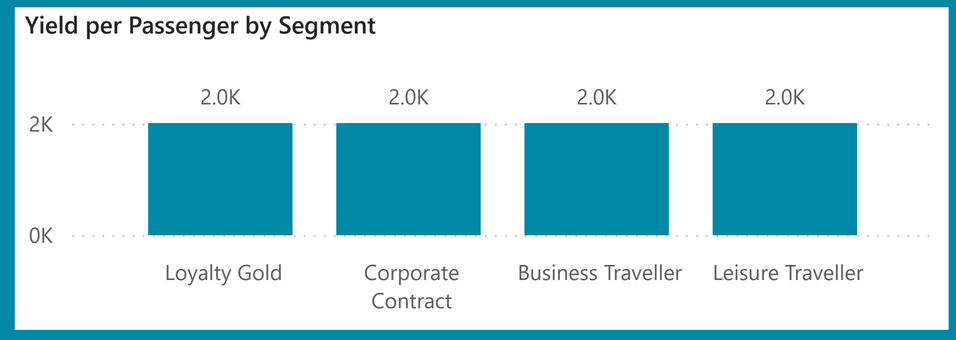


Revenue & Segment Mix Analysis:

Evaluates revenue concentration and premium mix distribution to highlight structurally higher-margin customer groups.

Booking Behavior & Yield Analysis

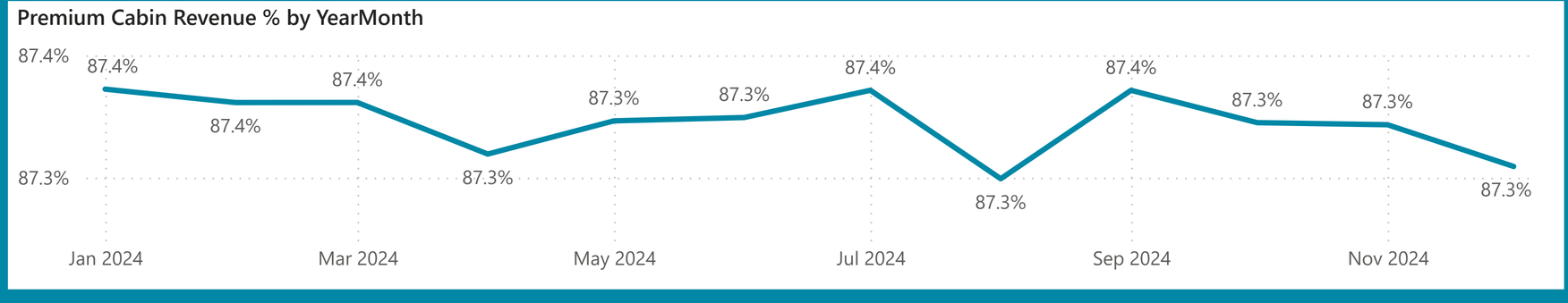




Booking Lead Days:

Reveals advance booking behavior differences across segments to assess demand visibility and forecasting reliability.

Premium Demand Trend



Premium Booking Trend:

Tracks evolution of premium revenue share across segments to detect long-term shifts in customer mix and profitability drivers.

OPERATIONAL INSIGHT (EXEC CONTROL VIEW) - YEAR 2022

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Select a Year:

- (Blank)
- 2022
- 2023
- 2024

Select a route:

- DXB-BOM
- DXB-JFK
- DXB-LHR
- DXB-SIN
- DXB-SYD

Select a fare class:

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  - Premium Eco.
  - Business
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- Standard

Select a segment:

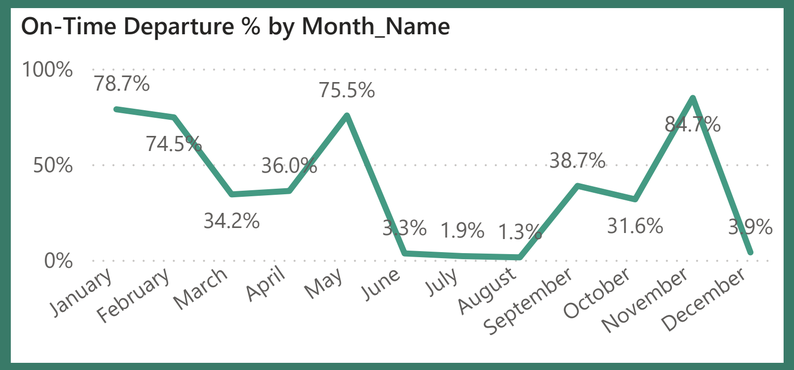
- Business Traveller
- Corporate Contract
- Leisure Traveller
- Loyalty Gold

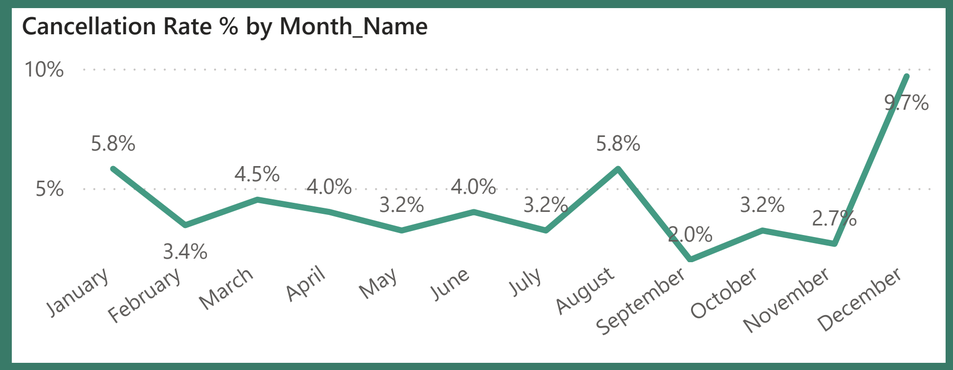
Operational Stability Snapshot

38.5%	4.3%	81.7%	2M	4.2bn
On-Time Departure %	Cancellation Rate %	Load Factor %	Total Passengers	Total Revenue

Operational Stability Snapshot:
Provides a consolidated view of execution reliability and disruption exposure to assess overall operational stability

Delay & Cancellation Trends

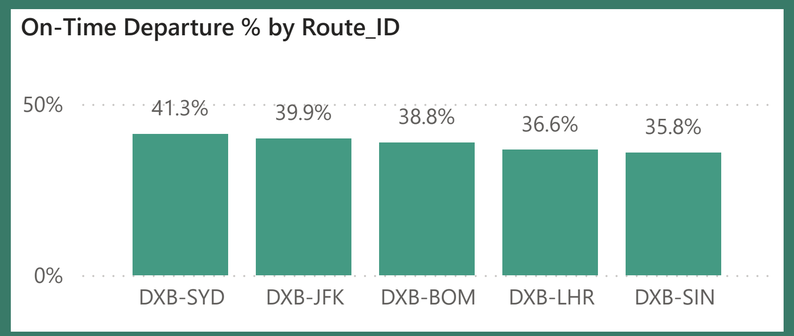


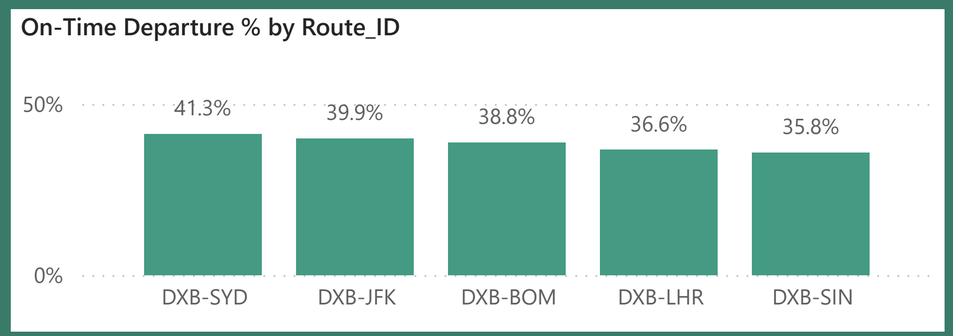


Delay & Cancellation Trends:

Reveals structural reliability patterns over time to distinguish temporary volatility from sustained operational weakness.

Operational Concetration

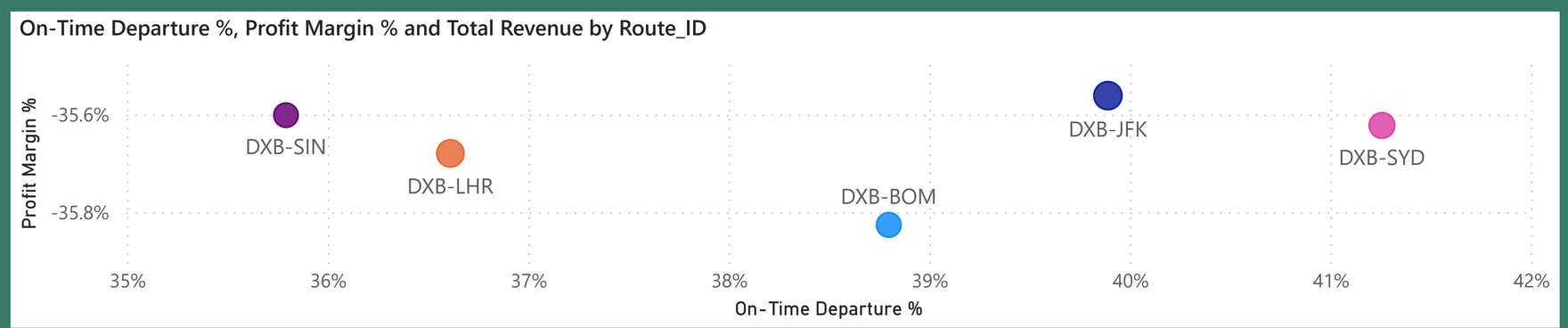




Operational Concentration:

Identifies routes or periods exhibiting disproportionate disruption, enabling targeted intervention rather than broad corrective measures.

Margin Impact Linkage



Linking Margin Impact:

Evaluates whether operational instability is materially compressing route-level profitability and elevating enterprise risk exposure.

KPI GOVERNANCE & ANALYTICS CoE

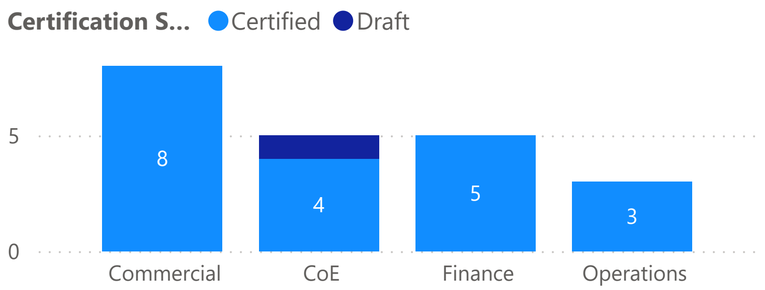
Governance Maturity Overview:

21	95.2%	100%	95%	95%
Total KPI	Certified KPI %	KPIs w/Assigned Ste...	KPIs w/Defined Quali...	SLA Compliance %

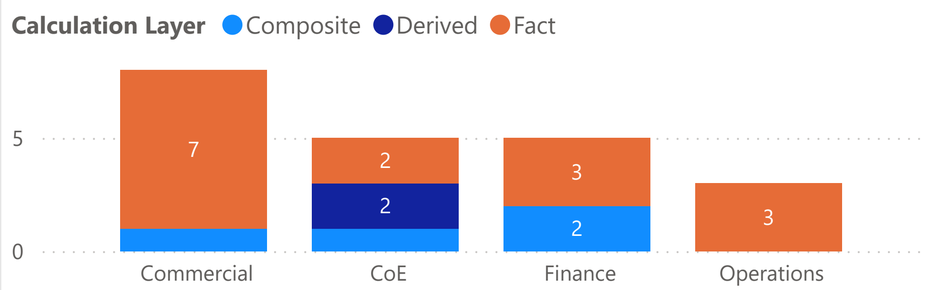
Governance & Coverage Snapshot:
Provides a consolidated view of governance maturity, certification coverage, and operational quality compliance across

Domain & Certification Control

Total KPI by Owner Function and Certification Status



Total KPI by Owner Function and Calculation Layer



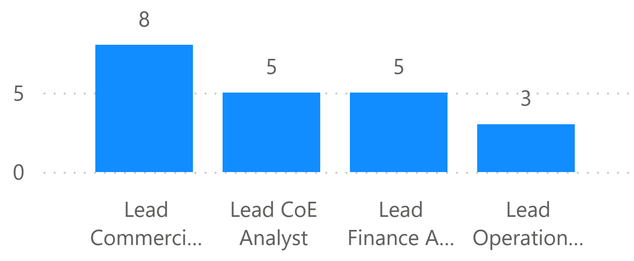
Domain & Certification Control:

Evaluates domain-level KPI ownership and certification discipline to ensure structural alignment and modeling transparency across the enterprise.

Ownership & Accountability Matrix

KPI Name	Owner Function	Data Steward	Certification Status	Calculation Layer
% Certified KPIs	CoE	Lead CoE Analyst	Certified	Derived
Advance Booking Wi...	Commercial	Lead Commercial ...	Certified	Fact
Ancillary Revenue %	Commercial	Lead Commercial ...	Certified	Fact
Cancellation Rate %	Operations	Lead Operations A...	Certified	Fact
Certified KPI Count	CoE	Lead CoE Analyst	Certified	Derived
Cost per Route	Finance	Lead Finance Anal...	Certified	Fact
Demand Momentu...	Commercial	Lead Commercial ...	Certified	Fact

Total KPI by Data_Steward

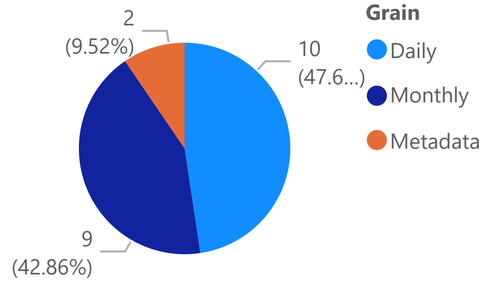


Ownership & Stewardship Accountability:

Surfaces executive accountability and operational stewardship responsibility to enforce semantic ownership and data governance discipline.

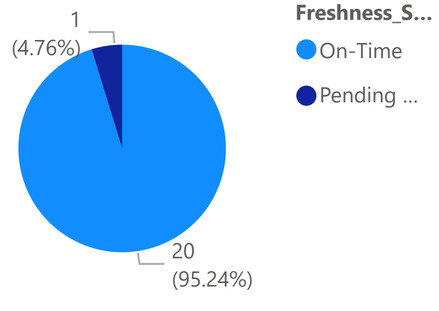
Data Quality, Lineage and Grain Integrity

Total KPI by Grain



KPI Name	Fact Source	Calc Layer	Fresh
% Certified KPIs	Dim_KPI_Definitions	Derived	As De
Advance Booking ...	Fact_Bookings	Fact	Daily
Ancillary Revenue...	Fact_Bookings	Fact	Daily
Cancellation Rate ...	Fact_Flights	Fact	Daily
Certified KPI Count	Dim_KPI_Definitions	Derived	As De
Cost per Route	Fact_Financials	Fact	Mont

Total KPI by Freshness_Status



Data Quality, Lineage and Grain Integrity:

Validates calculation grain alignment, refresh timeliness, and source lineage to prevent cross-domain ambiguity and reporting risk.